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Modelled health impacts of three stakeholder-selected policies to support healthy and environmentally sustainable population diets in New Zealand

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ABSTRACT

Food production and consumption impacts both planetary and population health. This research aimed to estimate the impacts of three food policies, selected by stakeholders, for the New Zealand (NZ) population: Exempting core, sustainable foods from Goods and Services Tax (GST), education about healthy sustainable foods and māra kai (food garden informed by mātauranga Māori) and community gardens. These were modelled using a proportional multistate lifetable model through to health gains (in Health Adjusted Life Years, HALYs), health equity impacts, and net costs (combining health system costs/cost savings and intervention costs in 2011 NZD). Changes in dietary intake were modelled through to changes in diet related disease incidence over the remaining life of the 2011 NZ population (n = 4.4 million, 3% discounted). Exempting core, sustainable foods from GST resulted in 87,410 HALYs (95% Uncertainty Interval (UI): 72,280 to 104,110) and health system cost savings of $927.3 Million (M), UI: (686.6 to 1216.6 M). Education about healthy sustainable foods generated small health impacts (1,170 HALYs, UI: 930 to 1,430) and health system cost savings ($9.6 M, UI: $6.8 to $12.9 M). Māra kai and community gardens generated 8,040 HALYs (UI: 4,720 to 11,800) and would cost the government $364.3 M (UI: $267.6 to $464.6 M). All scenarios produced more age-standardised per capita health gain for Māori (Indigenous population) than non-Māori. The first two policies were cost saving while the māra kai and community gardens policy was borderline cost-effective. This research shows that these policies have the potential to improve population health and could be used to inform food policy in high-income countries. For NZ, these policies could also reduce ethnic health inequities; careful policy design involving relevant stakeholders, including Māori communities, is needed to ensure health and equity benefits are realised.

1. Introduction

Dietary intake is an important risk factor for cardiovascular disease, diabetes and various cancers (Afshin et al., 2019), with an appreciable reduction in disease burden expected should population diets shift towards healthier options. High intakes of sodium and low intakes of wholegrains and fruits are the leading dietary risk factors for death and disability adjusted life-years (Afshin et al., 2019) with obesity due to poor diet leading to further incidence of chronic disease. NZ diets are low in fruit and vegetables and contribute to high rates of overweight and obesity (University of Otago and Ministry of Health, 2011, Ministry of Health, 2023).

NZ has a history of colonisation, resulting in health inequities experienced by Māori, the Indigenous population (Reid and Robson, 2000). This has impacted access to healthy foods for Māori. Māori are less likely to eat recommended servings of fresh fruit and vegetables, and are more likely to eat processed meat than the general population, dietary practices which are risk factors for non-communicable diseases (Afshin et al., 2019). These inequities could be reduced through policy change focused on ensuring equity of health status for Māori, working in partnership with Māori communities as guided by Te Tiriti o Waitangi (The Treaty of Waitangi, glossary of Māori words included in Supplementary Material) (Waitangi Tribunal, 2019, Public Health Advisory Committee, 2024).

Alongside the health consequences directly associated with poor diet, the health and health equity effects of environmental degradation

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are widely apparent and could pose extreme risks to human health through disruptive climate change and other environmental impacts (Ebi and Hess, 2020, Majeed and Ozturk, 2020, Romanello et al., 2022). The interrelated combination of obesity, undernutrition and climate change has been described as a global syndemic (Swinburn et al., 2019). In 2022, the global agrifood system was responsible for approximately 30% of all anthropogenic GHG emissions (FAO, 2024) with agriculture being responsible for over half of NZ's GHG emissions (Ministry for the Environment, 2024). Shifts in diet away from animal protein and processed foods and towards more vegetables, legumes, wholegrains and fruit are likely to have health and environmental co-benefits (Rockström et al., 2025).

The EAT-Lancet Commission proposed a 'Great Food Transformation' including a global reference diet to bring the six food system related earth system processes into safe operating boundaries (Willett et al., 2019). The reference diet is high in vegetables, legumes, wholegrains, fruit and plant sources of protein and low in animal sources of protein. The Commission takes a global perspective and notes that their framework does not provide a plan for translating these global targets to national targets or provide policy solutions for specific contexts. There exists, therefore, an important research gap on how individual countries can shift current population diets towards this healthier and more sustainable reference diet.

Fig. 1 illustrates the food system context in NZ, alongside the potential health and environmental benefits of shifts towards sustainable dietary intakes. Results from stakeholder consultation undertaken within the current project (Kidd et al., 2025, 2026) outlined the importance of strong local food networks and being able to grow and collect food locally for New Zealanders, all of which impact on food security and food system resilience. This local food system sits within a wider national food system driven by export markets which can overshadow the importance of providing healthy, affordable and sustainable food for New Zealanders.

Due to the size of food system impacts and potential for these to be positive, multiple policies which improve human health and the environment should be implemented simultaneously. Policies should be evidence based, have stakeholder buy-in and be implemented, evaluated and adapted over the long term. Burgaz et al. (2024) outline a range of policies that improve obesity and non-communicable diseases (NCDs) alongside climate change and biodiversity loss. Policies listed both here and as WHO best buys (World Health Organization, 2024) include reformulation, front of pack and menu labelling, marketing, taxes, subsidies and public food procurement. Additionally, reformulation, front of pack labelling and mass media campaigns for healthy diets have been identified as quick buys for the prevention of NCDs (expected to demonstrate impact on population health within five years or less) (Galea et al., 2025). Research on the cost-effectiveness and scope of health gain for policies designed to support healthy sustainable consumption in NZ, a small high-income country, can help policy makers select appropriate strategies to implement, both in NZ and globally. This article aims to estimate the health, health system costs, and equity impacts of three stakeholder identified dietary policies to support New Zealanders to consume more healthy and sustainable foods.

2. Methods

2.1. Selection of modelled policies

Modelled policies were selected through a 3-stage process using qualitative interviews and focus groups with key stakeholders (Kidd et al., 2025, 2026). Six focus groups with a total of 57 stakeholders from government ministries, a civil society group recruited through community organisations and flyers and contributors from two Māori communities, one based in an urban area and one based more rurally. Interviews were also carried out with 13 academic and food industry representatives. Stakeholders initially suggested a range of policies that could potentially help support New Zealanders to consume more healthy and environmentally sustainable food (N = 111). The policies that had direct evidence on the effect on dietary intake (specifically the dietary risk factors we model (Cleghorn et al., 2017)) were taken to the subsequent stage (N = 14). In this stage stakeholders and project advisors rated the policies on five criteria (improve Māori health, improve NZ diets, improve the environment, ease of implementation, acceptability). Average ratings were considered in the whole group and in Māori participants specifically. Top rated policies were taken through to the final stakeholder consultation. Here stakeholders discussed how best to implement these policies in the NZ context. The design of the five modelled policies were based on the details of these discussions. In this paper we present the three policies modelled in the whole population, excluding two aimed at children only. The proportions of the population



Fig. 1. Health, environmental and resilience co-benefits of a sustainable food system supporting sustainable dietary intake.

that are affected by these policies are presented in Fig. 2 and Table 1. The costs associated with implementing these policies are presented in Table 2.

2.2. Policy 1: goods and services tax (GST) exemption

Under the current GST system in NZ, almost all goods and services are taxed at the rate of 15%, this includes all sold food. The design of the current NZ system, with it's single rate and few exemptions, is intended to simplify the tax system and reduce administrative costs. This modelled policy simulated removal of GST from certain foods. Three scenarios were modelled under this policy:

- S1.1: GST exemption applied to core sustainable foods.
- S1.2: GST exemption applied to core foods.
- S1.3: GST exemption applied to fruits and vegetables.

Core foods were defined using the Australian definitions of core foods. To define 'core' NZ food groups, model food groups were matched to the core foods defined by the Australian system (Legal Database | ATO). Sustainable foods were defined as those food groups included in the EAT Lancet diet (Willett et al., 2019). To be included in S1.1 food groups needed to fit both categories. In practice, regulation would likely be needed to ensure retailers did not absorb the cost savings from these price reductions and savings would be passed onto consumers.

2.2.1. Population dietary intake data used in policy 1: GST exemption

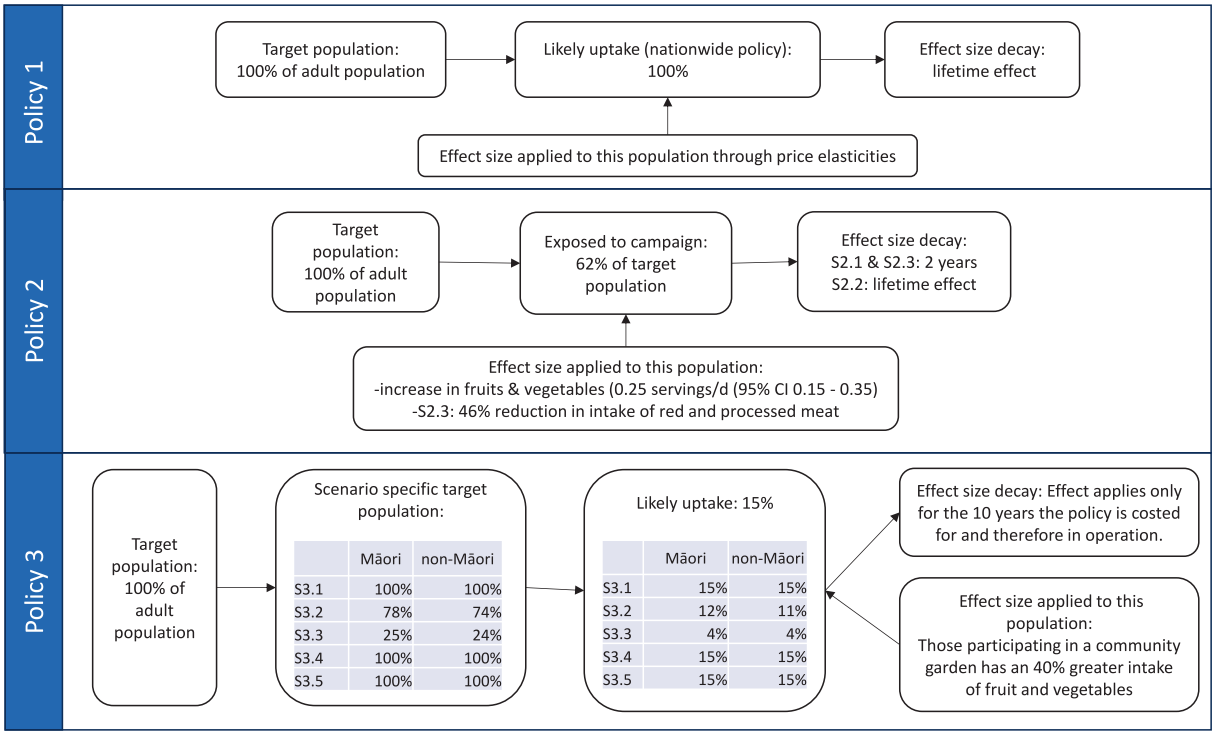
Dietary intakes were sourced from 24-h dietary recalls reported in the most recent (2008/09) New Zealand Adult Nutrition Survey (NZANS, n = 4721 individuals aged 15 years or older). Baseline dietary intake of 346 food groups was used in modelling GST exemption. Changes in price were modelled through NZ specific price elasticities to estimate the resulting change in dietary intake. This average change in dietary intake was used to calculate the change in dietary risk factors by sex and ethnic (Māori and non-Māori) groups which was subsequently linked through to the diet proportional Multistate Lifetable model (pMSLT).

Initial price elasticities were from the SPEND Study, conducted for NZ (Nnoaham et al., 2009, Mhurchu et al., 2013). These are in a 24 by 24 matrix (see Fig. 2 in the technical report (Cleghorn et al., 2017)) of own- and cross-PEs (with standard errors for default uncertainty). These 24 food groups were matched to the 346 food groups used in the intervention model. This gives us 24 overall food groups and 338 food subgroups (ignoring 5 'alcoholic beverage' groups, 2 'dietary supplement' groups and 1 'not applicable' group). The 24 by 24 price elasticity matrix was expanded to a 338 by 338 matrix following the methodology outlined in the technical report (Cleghorn et al., 2017).

2.3. Policy 2: education about healthy sustainable foods

This modelled policy was a mass media campaign which would encourage consumption of healthy, sustainable foods in the general population. Such a policy could utilise TV advertising, online, radio and print media. This would (likely) be led by the Ministry of Health (with input from iwi and other relevant government agencies such as Ministry for the Environment and Ministry for Primary Industries). Alongside a focus on healthy eating, it would include key topics such as how Aotearoa NZ grows food, how food production practices affect people and the environment and promoting meat-free meals (e.g. at least once a week). Input to the design of such a mass media campaign by Māori would be essential to ensure this policy does not contribute to dietary and health inequities. Three scenarios were modelled under this policy:

- S2.1: Assume the increase in fruit and vegetable intake is maintained for two years total.
- S2.2: Assume the effect on fruit and vegetable intake is maintained for the lifetime of the cohort.
- S2.3: S2.1 plus a reduction of red and processed meat.



	Māori	non-Māori
S3.1	100%	100%
S3.2	78%	74%
S3.3	25%	24%
S3.4	100%	100%
S3.5	100%	100%

	Māori	non-Māori
S3.1	15%	15%
S3.2	12%	11%
S3.3	4%	4%
S3.4	15%	15%
S3.5	15%	15%

Fig. 2. Proportion of the population affected by all policies (S2.1: Effect size for 2 years, S2.2: Effect size for lifetime of cohort, S2.3: S2.1 plus effect on red and processed meat intake, S3.1: All marae and suburbs in NZ (0.5 FTE/garden), S3.2: All marae and matched number of suburbs in NZ (0.5 FTE/garden), S3.3: 1/3 of Marae and matched number of suburbs in NZ (0.5 FTE/garden), S3.4: All marae and suburbs in NZ (1 FTE/garden), S3.5: All marae and suburbs in NZ (1 FTE/15 garden)).

Table 1
Intervention parameters for the three policies.

	Policy 1: GST exemption (Time horizon: lifetime of cohort)	Policy 2: Education about healthy sustainable foods	Policy 3: Māra kai and community gardens (Time horizon: 10 years)
Scenarios	S1.1 Applied to core sustainable foods. S1.2 Applied to core foods. S1.3 Applied to fruits and vegetables.	S2.1 Assume the effect on fruit and vegetable intake is maintained for 2 years total. S2.2 Assume the effect on fruit and vegetable intake is maintained for the lifetime of the cohort. S2.3 S2.1 plus a reduction of red and processed meat.	S3.1 All marae and suburbs in NZ (0.5 FTE community champion/garden). S3.2 All marae and the same number of suburbs in NZ (0.5 FTE community champion/garden). S3.3 1/3 of marae and the same number of suburbs in NZ (0.5 FTE community champion/garden). S3.4 All marae and suburbs in NZ (1 FTE community champion/garden). S3.5 All marae and suburbs in NZ (1 FTE community champion/15 gardens).
Targeting of policy	Targets the whole of the NZ adult population.	62% of adult population exposed to the campaign (average of (Morley et al., 2019; 2022)).	Target population varied depending on the number of marae and suburbs having a garden. 15% of those with potential access to māra kai/community gardens. This was based on survey carried out in areas with community gardens in the USA. 15% of respondents reported that they or a member of their household had participated in a community gardening project in the last 12 months (Alaimo et al., 2008).
Effect size	A price reduction equivalent to the removal of GST (−13%) was applied to the price of food groups in the model categorised as both core and sustainable (S1.1), core (S1.2) or fruit and vegetables (S1.3). To define 'core' food groups, model food groups were matched to the core foods defined by the Australian system (Legal Database | ATO). To define 'sustainable' food groups they were matched to the food groups included in the Eat Lancet commissions diet (Willett et al., 2019). These were then modelled through NZ price elasticities (Mhurchu et al., 2013). Full methods relating to price elasticities outlined in the model's technical report (Cleghorn et al., 2017).	Meta-analysis evidence for nutrition-based mass media campaigns shows a significant increase in fruits and vegetables (0.25 servings/d (95% CI 0.15–0.35) (Afshin et al., 2015). 46% reduction in intake of reduced red & processed meat based on a text message campaign (Carfora et al., 2017) (S2.3).	Those participating in a community garden have a 40% greater intake of fruits and vegetables (Alaimo et al., 2008).
Effect size decay	Assumed to be lifetime of cohort.	Effect on fruit and vegetable intake operating for 2 years total based on evidence of change in consumption a year after a similar campaign was run in Australia (Dixon et al., 1998) (S2.1 & S2.3). Assumed to be lifetime of cohort (S2.2).	Effect applies only for the 10 years the policy is costed for and therefore in operation.

GST: Goods and Services Tax; FTE: Full time equivalent.

There is a lack of evidence on the long-term effect of mass media campaigns on dietary intake so the first and last scenarios were only modelled to change dietary intake (Afshin et al., 2013, Afshin et al., 2015) in the short term (Dixon et al., 1998). The second scenario modelled a lifelong effect to illustrate the potential of a well-designed long-term mass media campaign. In the final scenario we modelled a 46% reduction in red and processed meat intake for a year of the cohort's lifetime. However, the evidence on the effect of a mass media campaign on red and processed meat intake is weak: participants were only followed up for 1 week (Carfora et al., 2017).

2.4. Policy 3: Māra kai (food garden informed by mātauranga Māori) and community gardens

This modelled policy would provide land, funding and a community champion to set up and maintain community gardens in suburbs and māra kai at or near marae around the country. For those marae and suburbs that already have māra kai or community gardens, the provided support could be used to help maintain and expand their gardens. Five scenarios, which varied the number of gardens and the level of support, were modelled under this policy:

- S3.1: Māra kai would be set up in all marae (N = 773) and community gardens in all suburbs (N = 1044) in NZ. A part-time (0.5 full-time equivalent, FTE) community champion would be paid for each garden.
- S3.2: Māra kai would be set up in all marae (N = 773) and an equivalent number of community gardens in some suburbs (N = 773) in NZ. A part-time (0.5 FTE) community champion would be paid for each garden.
- S3.3: Māra kai would be set up in a third of all marae (N = 250) and an equivalent number of community gardens in some suburbs (N = 250) in NZ. A part-time (0.5 FTE) community champion would be paid for each garden.

Table 2
Intervention costs for the three policies in 2011 NZD.

	Policy 1: GST exemption (Time horizon: lifetime of cohort)	Policy 2: Education about healthy sustainable foods (Time horizon: lifetime of cohort)	Policy 3: Māra kai and community gardens (Time horizon: 10 years)
Start-up costs	**$6.11 M** Cost of law: $3.45 M (Wilson et al., 2012). Further start-up costs were based on the Health Star Rating (HSR) labelling scheme operating in NZ: $2.66 M (CPI adjusted to 2011$ from the 2019 costs of $2.92 M) (MP Consulting 2019). These costs covered governance and operations of the scheme including monitoring, evaluation, social marketing and education.	**$684,271** One off campaign. Costs based on a similar campaign from Australia (Ananthapavan et al., 2022). Pre-campaign concept testing; Concept development and animatic creation: $41,790. Pre-campaign/production staff time costs; Advertising production; Creative agency fees; Web hosting and support: $642,481.	**Varied by scenario:** Grant for each garden at $2000/garden based on the existing Te Puni Kōkiri (Ministry of Māori development) funded programme (Flourishing Māra kai (tpk.govt.nz)). Additional $700 for gardens requiring new soil (an estimated 24% of gardens (Strawbridge 2022)) • S3.1: 1817 gardens: $3.94 M • S3.2: 1546 gardens: $3.35 M • S3.3: 500 gardens: $1.08 M • S3.4: 1817 gardens: $3.94 M • S3.5: 1817 gardens: $3.94 M
Ongoing fixed costs	**$41.75M** Covers continued governance, operations, monitoring, evaluation, social marketing and education in NZ. Costs based on annual spend on HSR scheme implemented in NZ: $379,591/yr for the lifetime of the cohort (110 years). (MP Consulting 2019).	**$579,572** Costs based on a similar campaign from Australia (Ananthapavan et al., 2022). These costs represent costs over the campaign broadcast: Advertisement broadcast (television, radio, cinema, online and search engine marketing, outdoor, press, Indigenous media); Media buying and planning fees; Monitoring fees; Broadcast supporting staff time costs; Electronic support materials.	**$13.42 M** Ongoing fixed costs calculated for 10 years of policy. 16 Regional co-ordinator staff costs based on Local Government Salary in NZ for a Project Manager: $83, 860 per annum (Project Manager, (Unspecified Type/General) Salary in New Zealand in 2026 | PayScale).
Ongoing variable costs	**$0M** Loss of GST revenue is not included in the policy costs.	**$0M**	**Varied by scenario:** Ongoing variable costs calculated for 10 years of policy.Community champion staff costs (living wage: $23.65/hour/staff) • S3.1: 0.5 FTE staff/garden (1817 gardens): $418.98 M • S3.2: 0.5 FTE staff/garden (1546 gardens): $356.49 M • S3.3: 0.5 FTE staff/garden (500 gardens): $115.29 M • S3.4: 1.0 FTE staff/garden (1817 gardens): $837.95 M • S3.5: 1.0 FTE staff/15 gardens (1817 gardens): $55.86 M
Total costs	**$47.86 M**	**$1.26 M**	**Varied by scenario:** • S3.1: $436.33 M • S3.2: $373.26 M • S3.3: $129.80 M • S3.4: $855.31 M • S3.5: $73.22 M

Costs have been Consumer Price Index (CPI) adjusted to the 2011 base year of the model and Purchasing Power Parity (PPP) adjusted from foreign currencies when required.
GST: Goods and Services Tax; FTE: Full time equivalent; HSR: Health Star Rating labelling scheme.

- S3.4: Māra kai would be set up in all marae (N = 773) and community gardens in all suburbs (N = 1044) in NZ. A full-time community champion would be paid for each garden.
- S3.5: Māra kai would be set up in all marae (N = 773) and community gardens in all suburbs (N = 1044) in NZ. A full-time community champion would be paid for, however they will be responsible for supporting 15 māra/gardens per person.

Robust funding to ensure longevity and centring community champions to help run the gardens would be important elements of this policy. The policy would be financed nationally and distributed to relevant local communities. This policy would require high-value land with good soil quality being allocated preferentially to the gardens. The gardens would incorporate food hubs with connections to existing farmers and gardens which would be co-ordinated by the paid community champions.

This policy would aim to build and help sustain existing māra/gardens and would centre community champions for the māra. Key considerations for māra kai include having mana motuhake, security and self-determination at the core, and access to māra marae for urban Māori. Education/knowledge transfers would be included alongside the gardens, such as the importance of gardening, how to cook and eat the food, and using māra kai as a space to tell Māori stories through kai.

2.5. *Proportional multistate lifetable model*

The change in dietary risk factors were modelled through to health adjusted life years (HALYs), health equity between Māori and non-Māori and health system costs or cost savings from diet-related diseases using an established diet pMSLT macrosimulation model (Cleghorn et al., 2017, Cleghorn et al., 2018).

The NZ population in 2011 (n = 4:4 million) were modelled out to death or until age 110 in the diet pMSLT model. A discount rate of 3% was used in the main analyses. The model is parameterized with rich national data by sex, age, and ethnicity. The diet pMSLT model includes a range of dietary risk factors (high intake of red meat, processed meat and sugar-sweetened beverages as well as low intake of fruit, vegetables, and nuts and seeds) and 8 diseases associated with one or more modelled dietary risk factors: coronary heart disease, stroke, type 2 diabetes, and multiple cancers (esophageal, colorectal, ovarian, head and neck, lung).

In this model the entire NZ population that was alive in 2011 is simulated over the remainder of their lives. The model consists of a main lifetable with projected all-cause mortality and morbidity rates and diet-related disease-life tables running in parallel. Disease incidence, case-fatality and remission (cancers only) determine the proportion of the population in each parallel disease state. Changes in dietary risk factors resulting from the policies are combined with disease-specific relative risks obtained from the Global Burden of Disease (GBD) study (Forouzanfar et al., 2015) through population impact fractions that alter the incidence of diet-related diseases in the model. The change in incidence of disease is modelled through to a reduction in morbidity and mortality from each disease which are combined to calculate HALYs. Morbidity is quantified for each disease as prevalent years of life lived with disability from the NZ Burden of Disease Study (Ministry of Health, 2013), divided by the number in the total population. Modelling assumes that the distributions of each modelled dietary risk factor can be treated as though independent of other risk factors. The incidence, remission, and case-fatality rates for each disease were modelled independently of other disease rates.

A health system perspective was taken for net costs which were calculated by combining intervention costs and health system costs/cost savings. The incremental cost effectiveness ratio (ICER) was calculated by dividing the net costs by the HALYs generated and represents the amount the government would need to pay for each HALY. The willingness-to-pay threshold was set to the level of gross domestic product per capita in 2011 (i.e. 2011 $NZ 45 000) per HALY gained. Sex- and age-specific health system costs (in NZ dollars, 2011) were calculated from individually-linked data for publicly-funded (and some privately-funded) health events occurring 2006–10 (Kvizhinadze et al., 2016). These included hospitalisation costs, inpatient procedures, outpatient costs, pharmaceutical costs, laboratory costs, and expected primary care usage costs. Costs were calculated for individuals without a diet-related disease and who were not in the last six months of their life (sex- and age-specific), as well as disease-specific excess costs for people in the first year of diagnosis, last six months of life if dying of the given disease, and otherwise prevalent cases of each disease. The modelled intervention costs are outlined in Table 2 and only include costs to government.

95% uncertainty intervals (95% UI) are generated by Monte Carlo simulations (n = 2000) to reflect uncertainty around the input parameters in the model. Uncertainty was modelled around the following parameters using the following distributions: Log-normal distributions were included around disease-specific incidence, prevalence, case fatality rates, remission rates and disability rates; normal distributions around baseline dietary intake and price elasticities used in policy 1, effect size in policy 2 and 3, disease trends and intervention effect time lags; uniform distributions around the theoretical minimum risk exposure levels; beta distributions around the proportion of the population exposed to policy 2 and 3 and the effect size in policy 2, S2.3; a gamma distribution was included around net costs.

Ethnic inequities in health were quantified through estimating per capita HALYs for Māori and non-Māori separately and presenting the ratio of these HALY gains between Māori and non-Māori. Ratios over 1.0 indicated that, per person, Māori received more health gain than non-Māori for the policy. It was assumed that the targeting of policies (except for policy 3: Māra kai and community gardens which varied by ethnicity), effect size and effect size decay were the same for Māori and non-Māori. Population size, demographics, dietary intake, background mortality and morbidity and diseases rates differed between Māori and non-Māori.

2.6. Additional scenario analyses

In the main analyses the Māori population are disadvantaged due to higher background morbidity and mortality rates, i.e. there is a lesser "envelope" for potential health gains. To remove this disadvantage, we present an equity analysis (McLeod et al., 2014) where we set Māori background morbidity and mortality rates at the same level as non-Māori values. Each scenario was also run with no discounting (Supplementary Material, Supplementary Table 1). The standard discounting rate of 3%, used in the main analyses, reduces the value of the health gains experienced by future generations by 3% for each year into the future while undiscounted analyses give equal importance to the health of future generations as current generations. This approach is particularly important when viewing policy benefits through a sustainability lens, which aims to balance the health of current and future generations.

3. Results

3.1. Policy 1: GST exemption

Scenario 1.1, GST exemption on foods that are categorised as both core and sustainable results in modest increases in fruit and vegetable intakes (15 g and 28 g respectively), with very small decreases in red and processed meat and sugar sweetened beverages (SSBs, Table 3). GST exemption on core foods only (S1.2) followed a similar pattern but with smaller changes in fruit and vegetables, small increases in meat and a larger decrease in SSBs (7 g). GST exemption on fruit and vegetables only (S1.3) also resulted in modest increases in fruit and vegetables, 17 g

Table 3
Change in dietary risk factors in grams/day (percentage of baseline intake).

	Δ Fruit	Δ Vegetables	Δ Red meat	Δ Processed meat	ΔSSB	Δ Nuts and seeds
Policy 1: GST exemption						
S1.1	14.9 (10%)	28.0 (18%)	−2.0 (−4%)	−0.5 (−1%)	−2.2 (−2%)	0.1 (2%)
S1.2	7.7 (5%)	19.9 (13%)	1.0 (2%)	0.2 (%)	−6.8 (−6%)	−0.1 (−2%)
S1.3	16.5 (11%)	27.1 (18%)	−1.3 (−3%)	−0.7 (−1%)	−1.5 (−1%)	−0.2 (−3%)
Policy 2: Education about healthy sustainable foods						
S2.1	6.2 (4%)	6.2 (4%)	–	–	–	–
S2.2	6.2 (4%)	6.2 (4%)	–	–	–	–
S2.3	6.2 (4%)	6.2 (4%)	−15.2 (−29%)	−18.4 (−29%)	–	–
Policy 3: Māra kai and community gardens						
S3.1	8.8 (6%)	8.9 (6%)	–	–	–	–
S3.2	6.8 (5%)	6.9 (5%)	–	–	–	–
S3.3	2.1 (1%)	2.2 (1%)	–	–	–	–
S3.4	8.7 (6%)	8.8 (6%)	–	–	–	–
S3.5	9.1 (6%)	9.2 (6%)	–	–	–	–

SSB: Sugar sweetened beverages, GST: Goods and Services Tax.

and 27 g respectively, very small decreases in red and processed meat of 1 g each and of SSBs (2 g). Nuts and seed intakes did not change more than 0.5 g under any scenario.

All scenarios for this policy are cost saving to the health system (Table 4). S1.1 generated the largest health impacts (87,410, 95% Uncertainty Interval (UI): 72,280 to 104,110) and health system cost savings ($$927.3 Million (M), UI: $686.6 to 1216.6 M). The next largest health gain (82,410 (68,260 to 97,970)) and health system cost savings ($826.0 M (UI: $609.6 to $1086.3 M)) are seen in the fruit and vegetable only scenario (S1.3). Undiscounted heath gain and health system cost savings range from 329,570 and $2,108.4 M in S1.1, targeting core, sustainable foods, down to 140,270 and $386.4 M in S1.2, targeting core foods only (Supplementary Material, Supplementary Table 1).

The ratio of age standardised per capita health gain between Māori and non– Māori range from 1.3 (S1.3) to 2.1 (S1.2). For S1.1, health gain for Māori increases by 38% when the equity analysis was applied and the ratio of age standardised, per capita health gain for Māori increased to between 1.9 and 2.9 compared to non-Māori health gain.

3.2. Policy 2: education about healthy sustainable foods

All scenarios have an increase of 6.2 g (4% of baseline intake) of fruit and vegetables. S2.3 includes a reduction in red and processed meat of 15.2 g and 18.4 g (both 29% of baseline intake).

S2.1 generates small health impacts (1,170 HALYs, UI: 930 to 1,430)) and health system cost savings ($9.6 M, UI: $6.8 to $12.9 M). Extending the length of effect from 2 years total (S2.1) to assuming the change in fruit and vegetable intake lasts the cohorts lifetime increased health gain to 19,270 HALYs (UI: 15,550 to 23,700) and health system cost savings to $156.2 M (UI: $117.1 to $206.9 M). Combining a reduction in red and processed meat with the increase in fruit and vegetable intake (S2.3) give health gains for 9,360 HALYs (UI: 7,170 to 11,500) and health system cost savings of $165.0 M (UI: $122.1 to $217.1 M). Undiscounted heath gain was 2,170 in S2.1 where fruit and vegetable intake increased for 2 years, 73,910 in S2.2 when the effect lasts the cohorts lifetime, and 17,860 in S2.3, where the policy also decreases red and processed meat intake. Undiscounted health system cost savings follow the same pattern: S2.1: $8.7 M, S2.2: $282.1 M, S2.3: $225.6 M.

The ratio of age standardised per capita health gain between Māori and non– Māori was 1.5, 1.5 and 2.5 for S2.1, S2.2 and S2.3 respectively. In S2.1, health gain for Māori increases by 32% when the equity analysis is applied and the age standardised, per capita health gain for Māori increases from 1.5 times to 2.5 times that of non-Māori.

3.3. Policy 3: Māra kai and community gardens

S3.1, with māra kai for all marae and community gardens for all suburbs in NZ (for the subsequent 10 years and a part time (0.5 full time equivalent, FTE) community champion) has the largest increase in average fruit and vegetables (9 g of both). This increase is shared with scenarios S3.4 and S3.5 which also cover all marae and suburbs. The scenarios that cover a lower proportion of marae and suburbs have a lower average increase in fruit and vegetable intakes, approximately 7 g in S3.2 (All marae and a matched number of suburbs in NZ) and 2 g in S3.3 (1/3 of marae and matched number of suburbs in NZ).

The largest health gains were seen for all the scenarios targeting all marae and suburbs (S3.1, S3.4 and S3.5), between 8,000 and 8,040 HALYs. Specifically, for S3.1: 8,040 (UI: 4,720 to 11,800 HALYs) and health system costs of $364.3 M (UI: $267.6 to $464.6 M). S3.1 was borderline cost-effective with an Incremental Cost Effectiveness Ratio (ICER) of $45,300 per HALY gained (using 2011 $45,000, a governmental willingness-to-pay threshold set to the level of gross domestic product per capita per HALY gained).

Decreasing the number of marae and suburbs receiving this policy (S3.2 and S3.3) reduces both health gain and cost of the policy and the cost-effectiveness ratio increases (moves further away from being cost-effective) to $52,600 (S3.2) and $56,900 (S3.3) per HALY gained. S3.4, where one full time community champion was employed per garden was also not cost-effective with an ICER of $97,500/HALY gained and S3.5, where one community champion looked after 15 gardens, was cost-effective at $30 per HALY gained and only costing $200,000 over the life course of the cohort. Undiscounted heath gain ranged from 4,020 in S3.3 to 16,550 in S3.1, S3.4 and S3.5. Undiscounted health system costs ranged from a cost saving of $3.6 M in S3.5 with 1 full time community champion per 15 gardens to health system costs of $778.5 M in S3.4, with 1 full time community champion per garden.

The ratio of age standardised per capita health gain between Māori and non– Māori was 1.4 for all scenarios except S3.3 where the ratio was 1.7. In S3.1, health gain for Māori increases by 34% when the equity analysis is applied and the ratio of age standardised, per capita health gain for Māori compared to non-Māori changed from 1.4 to 1.9.

4. Discussion

This research modelled the likely health, health equity and health system cost impacts of three policies, which were informed by focus groups and interviews with a range of NZ stakeholders interested in

Table 4
Health Adjusted Life Years, health system costs, Incremental Cost-Effectiveness Ratios and age standardised per capita ratio of HALYs between Māori and non-Māori for all scenarios.

	Lifetime HALYs (Mean (95% UI))	Lifetime costs (2011 NZ$ million, Mean (95% UI))	ICER	Age standardised per capita ratio of HALYs between Māori and non-Māori
Policy 1: GST exemption				
S1.1	87,410 (72,280 to 104,110)	$-927.3 (−1216.6 to −686.6)	Cost saving	1.4
S1.2	38,140 (24,340 to 50,060)	$-244.9 (−562.5 to 184.4)	Cost saving	2.1
S1.3	82,410 (68,260 to 97,970)	$-826.0 (−1086.3 to −609.6)	Cost saving	1.3
Policy 2: Education about healthy sustainable foods				
S2.1	1,170 (930 to 1,430)	$-9.6 (−12.9 to −6.8)	Cost saving	1.8
S2.2	19,270 (15,550 to 23,700)	$-156.2 (−206.9 to −117.1)	Cost saving	1.5
S2.3	9,360 (7,170 to 11,500)	$-165.0 (−217.1 to −122.1)	Cost saving	2.4
Policy 3: Māra kai and community gardens				
S3.1	8,040 (4,720 to 11,800)	$364.3 (267.6 to 464.6)	$45,300	1.4
S3.2	6,030 (3,370 to 8,830)	$317.7 (238.5 to 408.3)	$52,600	1.4
S3.3	1,970 (1,170 to 2,880)	$111.8 (85.2 to 140.0)	$56,900	1.4
S3.4	8,030 (4,610 to 11,790)	$783.3 (606.2 to 968.7)	$97,500	1.3
S3.5	8,000 (4,480 to 11,800)	$0.2 (−47.2 to 40.3)	$30	1.4

HALY: Health Adjusted Life Years, ICER: Incremental Cost-Effectiveness Ratio.

sustainable healthy food systems and consumption. It found that the GST exemption and education about healthy sustainable foods scenarios were all cost-saving. The māra kai and community gardens scenarios varied from highly cost-effective if one full-time community champion was employed to manage 15 gardens (assuming this level of staffing allowed the community gardens to be maintained through volunteers), to not cost-effective if each garden had their own (either full-time or half-time) community champion. However, the scenario which covered the greatest number of gardens (S3.1) was borderline cost-effective.

Under the first policy, where core, sustainable foods are exempt from GST, moderate health gains and health system cost savings ($927 M) are seen. This modelling does not include the loss of income to the government due to the exemption. Using estimates of 2011 food costs and population numbers we would expect the revenue to the government to decrease by approximately $890 M a year. On balance, we expect all GST exemption scenarios to cost the government money if loss of GST income is considered.

GST exemption affects the whole population and is likely to cause small shifts in population dietary intake over their lifetimes to generate these health impacts. Under the assumption that Māori and non-Māori respond the same to these price changes, larger per capita health gains are generated for Māori suggesting this policy could reduce health inequities between Māori and non-Māori. Additionally, there is evidence that Māori are more sensitive to changes in food prices (Mhurchu et al., 2013) which may increase Māori health gain further. These results show that reducing the cost of core foods can have a positive effect on health equity. Other mechanisms to reduce the cost of core food such as increased competition should be explored. The GST exemption policy in Australia is generally attributed to multiple studies finding a 'healthy' diet is cheaper than an 'unhealthy' diets, however, healthy diets are still found to be unaffordable to marginalised groups such as people on low incomes, and for First Nations people (Dawson et al., 2025). In NZ, there are equity criticisms of this policy as higher income individuals are likely to receive a larger absolute reduction in their food expenditure. The health benefits of this policy will need to be considered alongside the barriers identified through earlier stages of this project: loss of government revenue; the perceived difficulty associated with defining core and sustainable foods and associated industry pushback; retailers not passing on these savings to the customer, and the stated equity concerns.

There is a lack of evidence on the long-term effect of mass media campaigns on dietary intake so the policy to educate the public on healthy sustainable foods was only modelled to impact dietary intake in the short term. As such the health gains were relatively small. Under the assumption that dietary intake would change for the lifetime of the cohort health gain increased but only to about a fifth of the GST exemption of core sustainable food policy. Modelling a 46% reduction in red and processed meat intake for a year of the cohort's lifetime generated 8 times higher health gain than a change in fruit and vegetable intake alone. However, the evidence on the effect of a mass media campaign on red and processed meat intake is weak: participants were only followed up for 1 week (Carfora et al., 2017). To ensure the relevance and effectiveness of this policy, stakeholders emphasised the importance of a clear definition for 'sustainable and healthy' food, suggested implementing this policy alongside other food policies, that the policy be led by the Ministry of Health with cross sector collaboration and that a separate framework for Māori be developed.

The scenarios under the policy to increase māra kai and community gardens varied the number of gardens and the intensity of involvement of community champions. The results show that the most extensive coverage gives the greatest health gain, as would be expected. It is the most expensive (to cover the costs associated with paying community champions) but it is also the scenario which is the closest to being cost-effective. Funding for needed resources and strong oversight and leadership to ensure longevity was emphasised by stakeholders when discussing this policy. They envisioned this policy as wider than just community gardens as they could provide spaces to tell Māori stories through kai, opportunities for knowledge transfers, they could act as food hubs and provide connections to local farmers. Barriers such as how to ensure equitable access to gardens and māra kai and how to ensure the longevity of the gardens need to be considered when designing these policies. For this policy in particular there would be wider benefits than the chronic disease reduction modelled. Relevant benefits, illustrated in Fig. 1, include strengthening local food networks, improved food security and food system resilience alongside benefits to social cohesion and wellbeing.

Although all policies are either cost saving or could be designed to be cost-effective, reductions to health system expenditure occur over a long time period (the lifetime of the 2011 cohort). For education about healthy sustainable foods and māra kai and community gardens these require up front expenditure and GST exemption will lose government revenue. Governments need long-term vision to value the benefits of such policies. High and rising rates of diet related disease need to be addressed. Implementing a variety of policies with both short- and long-term impacts on the food environment and food choices is key to reducing these disease rates.

All modelled policies generated more age standardised per capita health gains for Māori than non-Māori with policy 2, education about healthy sustainable foods, having the biggest difference in per capita health gain between Māori and non-Māori. There are a number of assumptions that would need to be met for these greater health gains to be realised, especially for policy 2 which requires careful design and targeting to ensure positive dietary change is either the same or greater for Māori. All policies should be designed by and with Māori to meet the assumptions around equal uptake and effectiveness for Māori.

Academic literature (Hammond and Myles, 2000, Atkinson et al., 2014, Iyer et al., 2025) debates the impact of discounting on intergenerational equity, where the health of current populations is essentially prioritised over that of future generations. These arguments are particularly pertinent when considering environmental impacts. This research focuses on policies that can improve both health and the environment and has presented both discounted (3%) and undiscounted results. The undiscounted health gain is between 1.9 and 3.8 times greater than the health gain discounted at 3% but the relative health gain between the different policies and scenarios does not change.

The modelled policies were selected by stakeholders due to their likely ability to improve NZ diets, Māori health, and the environment as well as being feasible and acceptable. We were not, however, able to model the impact on GHG emissions of all policies due to a lack of data on their likely impact on emissions. Policy 2 and 3 increased fruit and vegetable intake but there was no evidence of what foods this increase might displace (if any) in the diets. We therefore cannot comment on the climate impact of these policies, despite them being selected and designed to be environmentally sustainable.

Evidence of policy effectiveness, the length of effect and policy uptake were based on *meta*-analyses (Afshin et al., 2015), if available, or the best, most relevant international evidence available (Dixon et al., 1998, Alaimo et al., 2008, Carfora et al., 2017, Morley et al., 2019; 2022), except for the GST policy which was based on NZ price elasticities (Mhurchu et al., 2015). These intervention parameters were applied to NZ specific demographic, diet and disease data (Cleghorn et al., 2017). Countries with comparable populations and disease rates could use these results to inform their own policy design and especially to compare the relative benefits of policies. The results from the current analyses can be compared with other dietary policies published using the same dietary model including a range of food and beverage taxes (Blakely et al., 2020, Grout et al., 2022), a food tax based on GHG emissions (Cleghorn et al., 2022) and interventions targeting individual behavior change (Cleghorn et al., 2020, Jones et al., 2022).

To understand and model these policies more accurately, further research on their effectiveness, their long-term effects, and their impacts on total dietary intake are needed. Additionally, further research is

needed into the effect on dietary intake in the medium and long term of a mass media campaign specifically on sustainable healthy dietary intake. This research has shown that the three modelled policies have the potential to improve population health and reduce ethnic health inequities in NZ. Due to the complementary nature of these policies, they could be implemented simultaneously to maximise impact. Based on the results of the current study, we recommend GST exemption for core, sustainable foods, alongside support to marae and communities to establish mara kai and community gardens across the country. These policies could be supported by a comprehensive and well-maintained mass media campaign to promote these policies, increase their social acceptance, and encourage healthy sustainable eating. For NZ, careful design involving relevant stakeholders including Māori communities is needed to ensure health benefits are realised. More broadly, this paper is highly relevant to global food-related public policies as results can be used to better understand the relative health impacts these policies could have in other high-income countries.

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CRediT authorship contribution statement

Christine Cleghorn: Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing. **Christina McKerchar:** Writing – review & editing, Writing – original draft, Methodology, Investigation. **Bruce Kidd:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis. **Cliona Ni Mhurchu:** Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.foodpol.2026.103049.

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